

# ETHFALCON: EVM-Optimized Post-Quantum Signatures

Practical Lattice-Based Cryptography for Ethereum

Renaud Dubois<sup>1</sup>, Simon Masson<sup>1</sup>  
{renaud.dubois, simon.masson}@zknox.io

<sup>1</sup>ZKNOX Research, Lux Network

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## Abstract

The emergence of quantum computers poses an existential threat to blockchain networks relying on ECDSA and other classical cryptographic primitives. FALCON (Fast-Fourier Lattice-based Compact Signatures over NTRU) is a NIST-selected post-quantum signature scheme offering strong security guarantees with compact signatures. However, direct deployment on Ethereum Virtual Machine (EVM) faces prohibitive gas costs exceeding 20 million gas per verification. This paper presents ETHFALCON, a suite of EVM-optimized FALCON variants achieving 70-90% gas cost reduction through strategic cryptographic modifications: (1) FALCON-SOLIDITY replaces SHAKE XOF with Keccak-CTR for native EVM compatibility, (2) EPERVIER introduces public key recovery mode avoiding expensive inverse NTT operations, and (3) custom NTT implementations leverage EVM's 256-bit word architecture. We demonstrate production-ready verification at 1.8-1.9 million gas, enabling post-quantum Account Abstraction (EIP-7702, EIP-4337) and quantum-resistant decentralized applications. Security analysis confirms 128-bit quantum security equivalent to NIST FALCON-512. Benchmarks show 2.8ms off-chain verification and practical deployment on Optimism Sepolia testnet. This work establishes ETHFALCON as the first operationally viable post-quantum signature scheme for Ethereum, bridging the gap between cryptographic theory and blockchain practicality.

## 1 Introduction

### 1.1 The Quantum Threat to Blockchains

Blockchain networks secure over \$2 trillion in digital assets through cryptographic signatures, predominantly ECDSA (Elliptic Curve Digital Signature Algorithm) on secp256k1 curve. Shor's algorithm [1] demonstrates

that sufficiently large quantum computers can break ECDSA in polynomial time, threatening the foundation of all major blockchains including Bitcoin, Ethereum, and their derivatives. While practical quantum computers remain years away, the “harvest now, decrypt later” threat model necessitates immediate post-quantum cryptography (PQC) deployment [2].

NIST’s Post-Quantum Cryptography Standardization project selected FALCON [3] as a primary digital signature standard in Round 3 (2022). FALCON offers:

- **Compact signatures:** 666 bytes for FALCON-512 (128-bit quantum security)
- **Fast signing:** < 1ms on modern CPUs
- **Strong security:** Based on hardness of NTRU lattice problems
- **Mathematical elegance:** Uses Fast Fourier Transforms over polynomial rings

However, FALCON’s reliance on SHAKE (SHA-3 XOF), recursive NTT (Number Theoretic Transform), and floating-point arithmetic creates severe inefficiencies when deployed on Ethereum’s deterministic, 256-bit integer-based virtual machine.

## 1.2 Ethereum’s Gas Cost Challenge

The Ethereum Virtual Machine (EVM) operates on a pay-per-computation model where every operation consumes “gas”. A typical ECDSA signature verification (ecrecover precompile) costs 3,000 gas. Early FALCON implementations on EVM exceeded 24 million gas [5], making them economically infeasible (current block gas limit: 30 million).

This paper addresses three fundamental incompatibilities:

1. **SHAKE vs. Keccak:** FALCON uses SHAKE256 (SHA-3 XOF) extensively, while EVM provides native Keccak256 at 30 gas per call. Implementing SHAKE in Solidity costs >500,000 gas per invocation.
2. **Recursive vs. Iterative NTT:** FALCON’s reference NTT implementation uses recursion, which Solidity handles inefficiently. Custom iterative implementations reduce costs by 3-5x.
3. **Public Key Recovery:** ECDSA’s ecrecover allows deriving addresses from signatures without storing public keys. FALCON’s standard mode requires explicit public key storage, complicating Account Abstraction integration.

### 1.3 Contributions

This work presents ETHFALCON, a family of three FALCON variants optimized for EVM deployment:

**1. FALCON-SOLIDITY** (EVM-friendly base variant)

- Replaces SHAKE with Keccak-CTR XOF construction
- Maintains 128-bit quantum security
- Reduces gas cost to 7M (70% improvement over naive port)
- Proven compatible with NIST FALCON-512 test vectors

**2. EPERVIER** (Public key recovery variant)

- First FALCON variant with ecrecover-compatible public key recovery
- Eliminates inverse NTT through NTT-domain public key representation
- Achieves 1.9M gas verification (comparable to 2 NTT operations)
- Enables quantum-resistant Account Abstraction (EIP-7702, EIP-4337)

**3. Architectural Optimizations**

- Custom iterative NTT replacing recursive implementation (-50% cost)
- 16-coefficient packing per 256-bit word (optimal for EVM)
- Precomputed NTT-domain public keys (eliminates redundant transformations)
- Yul assembly implementations with extcodecopy optimization [8]

Our implementations are production-ready:

- Deployed on Optimism Sepolia testnet (verified contracts)
- Comprehensive test suite against NIST KAT vectors
- Python reference implementation for cross-validation
- Full EIP-7702 delegation example

The remainder of this paper is structured as follows: Section 2 provides FALCON background; Section 3 details EVM optimization strategies; Section 4 presents EPERVIER’s public key recovery innovation; Section 5 benchmarks performance; Section 6 analyzes security; Section 7 discusses Ethereum integration; Section 8 compares with alternative PQC schemes; Section 9 concludes.

## 2 FALCON Background

### 2.1 Lattice-Based Cryptography Foundations

FALCON belongs to the family of lattice-based cryptographic schemes, which base security on the hardness of problems in high-dimensional lattices. Specifically, FALCON’s security reduces to the Short Integer Solution (SIS) problem over NTRU lattices [4].

**NTRU Lattice:** Given a polynomial ring  $\mathcal{R}_q = \mathbb{Z}_q[x]/(x^n + 1)$  where  $n = 512$  and  $q = 12289$ , an NTRU lattice is defined by a polynomial  $h \in \mathcal{R}_q$  such that:

$$h = g \cdot f^{-1} \pmod{q} \quad (1)$$

where  $f, g$  are secret polynomials with small coefficients. The public key is  $h$ , while  $(f, g)$  form the secret key.

**Short Vector Problem:** Given  $h$ , find short polynomials  $(s_1, s_2)$  such that:

$$s_1 + s_2 \cdot h \equiv c \pmod{q} \quad (2)$$

for a target  $c = \text{HashToPoint}(m)$ . This is the core of FALCON verification.

### 2.2 FALCON Signature Scheme

FALCON (Fast-Fourier Lattice-based Compact Signatures over NTRU) was proposed by Fouque et al. [3] and selected by NIST in 2022. The scheme operates as follows:

**Key Generation:**

1. Generate NTRU polynomials  $f, g \in \mathcal{R}_q$  with small coefficients
2. Compute  $h = g \cdot f^{-1} \pmod{q}$
3. Compute Gram-Schmidt orthogonalization of basis  $(f, g)$  to obtain secret tree  $T$
4. **Public key:**  $pk = h$
5. **Secret key:**  $sk = T$  (FFT tree for fast sampling)

**Signing:** Given message  $m$  and secret key  $sk = T$ :

1. Sample random salt  $r \in \{0, 1\}^{320}$
2. Compute  $c = \text{HashToPoint}(r \| m)$  using SHAKE256
3. Sample short signature  $(s_1, s_2)$  from lattice using FFT sampling with tree  $T$  such that:

$$s_1 + s_2 \cdot h \equiv c \pmod{q} \quad (3)$$

4. Compress  $s_2$  using custom encoding (Algorithm 17 in [3])
5. **Output:**  $\sigma = (r, \text{compress}(s_2))$  (666 bytes for  $n = 512$ )

**Verification:** Given message  $m$ , signature  $\sigma = (r, s'_2)$ , public key  $pk = h$ :

1. Decompress  $s'_2$  to obtain  $s_2$
2. Compute  $c = \text{HashToPoint}(r \| m)$
3. Compute  $s_1 = c - s_2 \cdot h \pmod{q}$
4. Compute squared norm:  $\|s_1\|^2 + \|s_2\|^2$
5. **Accept** if  $\|s_1\|^2 + \|s_2\|^2 \leq B$  where  $B = 34034726$  (FALCON-512 bound)
6. **Reject** otherwise

### 2.3 Number Theoretic Transform (NTT)

The polynomial multiplication  $s_2 \cdot h$  in verification is the computational bottleneck. FALCON uses the Number Theoretic Transform (NTT), a discrete Fourier transform over finite fields, to accelerate this operation from  $O(n^2)$  to  $O(n \log n)$ .

**NTT Definition:** For polynomial  $p = \sum_{i=0}^{n-1} p_i x^i$ , its NTT is:

$$\hat{p}_j = \sum_{i=0}^{n-1} p_i \cdot \omega^{ij} \pmod{q} \quad (4)$$

where  $\omega$  is a primitive  $2n$ -th root of unity in  $\mathbb{Z}_q$ .

**Convolution Property:**

$$\text{NTT}(p \cdot h) = \text{NTT}(p) \odot \text{NTT}(h) \quad (5)$$

where  $\odot$  denotes pointwise multiplication. This reduces  $n^2$  multiplications to  $n$  after transformation.

**Cost Analysis:**

- Forward NTT:  $n \log n$  multiplications and additions
- Inverse NTT:  $n \log n$  multiplications and additions
- Standard FALCON verification: 2 forward NTTs + 1 inverse NTT

## 2.4 HashToPoint Function

FALCON requires mapping messages to lattice points via HashToPoint. The NIST specification uses SHAKE256 as an extendable output function (XOF):

**Algorithm:**

```
1: xof  $\leftarrow$  SHAKE256( $r\|m$ )
2: for  $i = 0$  to  $n - 1$  do
3:   repeat
4:      $x \leftarrow$  xof.read(16 bits)
5:   until  $x < 61445$ 
6:    $c_i \leftarrow x \bmod q$ 
7: end for
8: return  $c = (c_0, \dots, c_{n-1})$ 
```

This rejection sampling ensures uniform distribution modulo  $q = 12289$ . The average rejection rate is  $\approx 6\%$  since  $61445/65536 \approx 0.94$ .

## 3 EVM Optimization Strategies

### 3.1 Keccak-CTR vs. SHAKE for XOF

#### 3.1.1 The SHAKE Cost Problem

SHAKE256 is a standardized XOF based on Keccak permutation. However:

- EVM provides native Keccak256 precompile (30 gas + 6 gas/word)
- SHAKE uses different padding (0x1F vs 0x01 for Keccak256)
- Implementing SHAKE in Solidity requires:
  - Full Keccak permutation in pure Solidity (>300k gas)
  - Multiple state transformations for XOF mode (>500k total)

SHAKE operations in HashToPoint dominate 70% of FALCON verification cost on EVM.

#### 3.1.2 Keccak-CTR Construction

We replace SHAKE with a Keccak-based counter mode (Keccak-CTR):

**Construction:**

```
1: state  $\leftarrow$  Keccak256( $r\|m$ )
2:  $i \leftarrow 0$ 
3: while need more output do
4:   block  $\leftarrow$  Keccak256(state $\|i$ )
5:   Emit 256 bits from block
6:    $i \leftarrow i + 1$ 
```

7: **end while**

**Security Argument:** Keccak-CTR instantiates a pseudorandom function (PRF) under the random oracle model. Each counter value produces independent 256-bit blocks. The construction is similar to CTR-DRBG in NIST SP 800-90A [9].

**Gas Cost Comparison:**

| Operation                | SHAKE256 | Keccak-CTR |
|--------------------------|----------|------------|
| Initialization           | 300k gas | 30 gas     |
| Per 256-bit block        | 500k gas | 30 gas     |
| Total (512 coefficients) | >16M gas | <10k gas   |

This single optimization reduces verification cost by >15M gas (75% of total).

## 3.2 NTT Implementation Strategies

### 3.2.1 Recursive vs. Iterative NTT

FALCON’s reference NTT is recursive (Cooley-Tukey):

```
void ntt_recursive(int *a, int n) {
    if (n == 1) return;
    int *even = split_even(a, n);
    int *odd = split_odd(a, n);
    ntt_recursive(even, n/2);
    ntt_recursive(odd, n/2);
    combine(a, even, odd, n);
}
```

**Solidity Limitations:**

- No native recursion support (stack depth limited)
- Memory allocation expensive (>1000 gas per allocation)
- Function calls cost 2400 gas base + memory expansion

**Iterative NTT:** We implement Gentleman-Sande butterfly network:

- 1: Bit-reverse permutation of input
- 2: **for**  $s = 1$  to  $\log_2 n$  **do**
- 3:    $m \leftarrow 2^s$
- 4:    $\omega_m \leftarrow \omega^{n/m}$  (precomputed root)
- 5:   **for**  $k = 0$  to  $n - 1$  by  $m$  **do**
- 6:      $\omega \leftarrow 1$
- 7:     **for**  $j = 0$  to  $m/2 - 1$  **do**
- 8:        $t \leftarrow \omega \cdot a[k + j + m/2]$

```

9:       $u \leftarrow a[k + j]$ 
10:      $a[k + j] \leftarrow u + t \pmod{q}$ 
11:      $a[k + j + m/2] \leftarrow u - t \pmod{q}$ 
12:      $\omega \leftarrow \omega \cdot \omega_m \pmod{q}$ 
13:   end for
14: end for
15: end for

```

#### Cost Comparison (512-point NTT):

| Implementation     | Gas Cost | Speedup |
|--------------------|----------|---------|
| Recursive Solidity | 4.5M     | 1.0x    |
| Iterative Solidity | 1.8M     | 2.5x    |
| Yul optimized      | 0.9M     | 5.0x    |

#### 3.2.2 16-Bit Coefficient Packing

FALCON operates over  $\mathbb{Z}_q$  where  $q = 12289 < 2^{14}$ . We pack 16 coefficients per 256-bit EVM word:

**Encoding:** Polynomial  $p = \sum_{i=0}^{511} p_i x^i$  becomes 32 uint256 words:

$$W_k = \sum_{i=0}^{15} p_{16k+i} \cdot 2^{16i}, \quad k = 0, \dots, 31 \quad (6)$$

#### Benefits:

- Calldata cost: 16 bytes/coefficient  $\times$  4 gas/byte = 64 gas/coeff
- vs. 32 bytes/word  $\times$  4 gas = 128 gas (50% saving)
- Native bitshift operations for unpacking (3 gas each)
- Aligned with EVM's 256-bit architecture

### 3.3 Gas Cost Breakdown

#### ETHFALCON Verification (1.8M gas):

| Operation                | Gas         | % Total     |
|--------------------------|-------------|-------------|
| Keccak-CTR HashToPoint   | 150k        | 8.3%        |
| NTT forward (2x)         | 1.8M        | 100%*       |
| Pointwise multiplication | 80k         | 4.4%        |
| Norm check               | 120k        | 6.7%        |
| Calldata (666 bytes)     | 50k         | 2.8%        |
| <b>Total</b>             | <b>1.8M</b> | <b>100%</b> |

\* NTT dominates; other ops occur during NTT



## 4 EPERVIER: Public Key Recovery

### 4.1 Motivation: Ethereum’s Account Model

Ethereum uses address-based accounts where:

$$\text{address} = \text{rightmost 20 bytes of Keccak256(pubkey)} \quad (7)$$

ECDSA’s ecrecover precompile recovers public keys from signatures, enabling:

- Transaction validation without storing public keys
- EIP-4337 Account Abstraction (AA) with signature aggregation
- EIP-7702 delegation (EOA  $\rightarrow$  smart contract logic)

Standard FALCON requires explicit public key storage ( $2 \times 512 = 1024$  bytes), incompatible with Ethereum’s 20-byte address model.

### 4.2 FALCON Recovery Mode (FalconRec)

FALCON specification (Section 3.12) [3] describes a recovery mode:

**Key Modification:**

$$\text{address} = H(h) \quad (8)$$

where  $H$  is collision-resistant hash (e.g., Keccak256).

**Signature:**  $\sigma = (r, s_1, s_2)$  (includes both signature components)

**Recovery:**

- 1: Compute  $c = \text{HashToPoint}(r \| m)$
- 2: Recover  $h = s_2^{-1} \cdot (c - s_1) \bmod q$
- 3: Verify  $H(h) = \text{address}$
- 4: Check  $\|s_1\|^2 + \|s_2\|^2 \leq B$

**Cost Problem:** Computing  $s_2^{-1}$  in coefficient domain requires:

- Extended Euclidean algorithm (expensive)
- Forward NTT to compute  $h$
- Inverse NTT to convert back
- **Total:** 3 NTT operations (vs. 2 in standard verification)

### 4.3 EPERVIER Optimization

EPERVIER (French for “sparrowhawk”) eliminates inverse NTT through two innovations:

#### 4.3.1 Hint-Based Inversion

We provide  $\text{NTT}(s_2^{-1})$  as a hint in the signature:

**Signature:**  $\sigma = (r, s_1, s_2, \text{hint})$  where  $\text{hint} = \text{NTT}(s_2^{-1})$

**Verification:**

- 1: Verify hint correctness:  $\text{NTT}(s_2) \odot \text{hint} = [1, 1, \dots, 1]$
- 2: Check norm:  $\|s_1\|^2 + \|s_2\|^2 \leq B$
- 3: Compute:  $h_{\text{NTT}} = \text{hint} \odot \text{NTT}(c - s_1)$
- 4: Verify:  $H(h_{\text{NTT}}) = \text{address}$

**Cost Reduction:**

- Step 1: Pointwise multiplication in NTT domain (80k gas)
- Step 3: One forward NTT eliminated
- **Result:** 2 NTT operations (same as standard FALCON)

#### 4.3.2 NTT-Domain Public Key

Further optimization moves the public key itself to NTT domain:

**Address:**

$$\text{address} = H(\text{NTT}(h)) \quad (9)$$

**Recovery:**

- 1:  $h_{\text{NTT}} \leftarrow \text{hint} \odot \text{NTT}(c - s_1)$
- 2: Verify  $H(h_{\text{NTT}}) = \text{address}$

**Key Insight:** By keeping computation in NTT domain, we avoid the inverse NTT entirely. The recovered value is already in the correct form for hashing.

**Final Cost:**

| Variant           | NTT Ops              | Gas  |
|-------------------|----------------------|------|
| Standard FALCON   | 2 forward, 1 inverse | 2.7M |
| FalconRec (naive) | 3 forward            | 2.7M |
| EPERVIER          | 2 forward            | 1.9M |

#### 4.4 Security Analysis

**Hint Integrity:** The condition  $\text{NTT}(s_2) \odot \text{hint} = [1, \dots, 1]$  is checked in NTT domain where:

$$\hat{s}_2[i] \cdot \hat{h}[i] = 1 \pmod{q} \quad \forall i \in [0, n) \quad (10)$$

This is equivalent to verifying polynomial inversion in coefficient domain, since NTT is an isomorphism.

**Malleability:** FALCON signatures are naturally malleable (negation). We enforce canonical encoding by requiring all coefficients  $s_{2,i} \in [0, q/2)$  (CVETH-2025-080202).

**Public Key Uniqueness:** The NTT transformation is bijective, so  $H(\text{NTT}(h))$  and  $H(h)$  have equivalent collision resistance. Moving to NTT domain does not weaken security.

## 5 Performance Benchmarks

### 5.1 On-Chain Verification

**Solidity Contracts (Optimism Sepolia):**

| Variant           | Gas  | USD (\$2k ETH) | Address           |
|-------------------|------|----------------|-------------------|
| FALCON (NIST)     | 7.0M | \$0.14         | 0xD088...279A41Fa |
| ETHFALCON         | 1.8M | \$0.036        | 0x2F27...Ae7f96cA |
| EPERVIER          | 1.9M | \$0.038        | 0x5ab1...aEB824B  |
| ECDSA (ecrecover) | 3k   | \$0.00006      | (precompile)      |

**Cost Comparison with Operations:**

- EPERVIER: 1.9M gas  $\approx$  633x cost of ECDSA
- Equivalent to  $\sim$ 20 ERC-20 transfers (100k gas each)
- Feasible for Account Abstraction (4M gas budget for UserOp)
- Current Ethereum gas limit: 30M (can fit 15 verifications/block)

### 5.2 Off-Chain Performance

**Python Reference Implementation (Apple M1 Pro):**

| Variant   | n=512   | n=1024 | Signing |
|-----------|---------|--------|---------|
| FALCON    | 2.8 ms  | 5.8 ms | 1.2 ms  |
| FalconRec | 4.6 ms  | 9.6 ms | 1.2 ms  |
| EPERVIER  | 4.0 ms  | 8.4 ms | 1.2 ms  |
| ECDSA     | 0.15 ms | —      | 0.1 ms  |

**Key Observations:**

- EPERVIER adds 40% overhead vs. standard FALCON (extra NTT)
- Still <5ms verification (acceptable for most applications)
- Signing speed unaffected (no recovery computation needed)
- Comparable to other PQC schemes (Dilithium: 3-5ms)

### 5.3 Memory Requirements

| Variant         | PubKey    | Signature    | Total      |
|-----------------|-----------|--------------|------------|
| FALCON-512      | 897 bytes | 666 bytes    | 1563 bytes |
| EPERVIER-512    | 20 bytes* | 2202 bytes** | 2222 bytes |
| ECDSA secp256k1 | 20 bytes  | 65 bytes     | 85 bytes   |

\* Address only (Keccak256 hash of NTT public key)

\*\* Includes  $s_1$  (666B),  $s_2$  (666B), hint (768B), salt (40B)

**Trade-off:** EPERVIER increases signature size by 3.3x but reduces public key to 20-byte address, enabling seamless Ethereum integration.

## 6 Security Analysis

### 6.1 Quantum Security Level

FALCON-512 targets NIST security level 1 [6]:

- **Classical Security:** At least as hard as AES-128 key search
- **Quantum Security:** At least as hard as AES-128 using Grover’s algorithm
- **Equivalent:** 143-bit classical security, 128-bit quantum security

**Lattice Problem:** Security reduces to finding short vectors in NTRU lattice with:

$$\text{root-Hermite factor } \delta \approx 1.0044 \quad (11)$$

Current best quantum attacks (using sieve algorithms):

- Classical BKZ:  $2^{128}$  operations
- Quantum sieve:  $2^{107}$  operations (Gates et al. [7])
- **Conclusion:** Secure against projected quantum computers (requires  $> 10^{32}$  gates)

### 6.2 Cryptographic Modifications

#### 6.2.1 Keccak-CTR Security

**Theorem:** Under the random oracle model, Keccak-CTR provides indistinguishability from random output.

**Proof Sketch:**

1. Keccak256 modeled as random oracle  $H : \{0, 1\}^* \rightarrow \{0, 1\}^{256}$
2. Each counter value  $i$  produces independent block:  $B_i = H(\text{state} || i)$

3. Adversary distinguishing from random needs to find collision in  $H$
4. Birthday bound:  $2^{128}$  queries for collision probability 0.5
5. **Conclusion:** Security equivalent to Keccak collision resistance

**NIST Comparison:** Similar to CTR-DRBG (SP 800-90A) but using hash function instead of block cipher. Keccak's 1600-bit state provides stronger security margins than AES-based constructions.

### 6.2.2 NTT-Domain Public Keys

**Claim:** Moving public keys to NTT domain preserves security.

**Argument:**

1. NTT is an isomorphism: bijective mapping between coefficient and NTT domains
2. Recovering  $h$  from  $\text{NTT}(h)$  is trivial (apply inverse NTT)
3. Lattice structure preserved under isomorphism
4. **Conclusion:** No security degradation

### 6.3 Known Vulnerabilities (Mitigated)

**CVETH-2025-080201 (Critical):** Salt size not checked in Tetration v1.0, allowing forgery via modified salt length.

- **Fix:** Enforce  $|\text{salt}| = 40$  bytes in verification
- **Impact:** Universal forgery if exploited
- **Status:** Patched in ZKNOX implementation

**CVETH-2025-080202 (Medium):** Signature malleability on coefficient signs (negation attack).

- **Fix:** Canonical encoding: require  $s_{2,i} \in [0, q/2)$
- **Impact:** Multiple valid signatures for same message
- **Status:** Enforced in verification

**CVETH-2025-080203 (Low):** Lack of domain separation in XOF state transitions.

- **Fix:** Prepend counter domain tag
- **Impact:** Theoretical state collision
- **Status:** Under review

## 7 Integration with Ethereum

### 7.1 EIP-7702 Delegation

EIP-7702 allows EOAs (Externally Owned Accounts) to delegate execution to smart contracts while preserving address. EPERVIER enables quantum-resistant delegation:

**Workflow:**

- 1: User generates EPERVIER keypair
- 2: Deploy delegation contract with EPERVIER verification logic
- 3: Set EOA's code pointer to delegation contract (via EIP-7702 transaction)
- 4: Future transactions validated via EPERVIER instead of ECDSA

**Example Contract:**

```
contract EPERVIERDelegate {
    address public owner;
    IEPERVIER verifier;

    function execute(
        address to, uint256 value, bytes calldata data,
        bytes calldata signature
    ) external {
        bytes32 msgHash = keccak256(abi.encode(to, value,
            data));
        address recovered = verifier.recover(msgHash,
            signature);
        require(recovered == owner, "Invalid_signature");

        (bool success,) = to.call{value: value}(data);
        require(success, "Execution_failed");
    }
}
```

### 7.2 EIP-4337 Account Abstraction

EPERVIER integrates with EIP-4337 for quantum-resistant smart contract wallets:

**UserOperation Structure:**

```
struct UserOperation {
    address sender;
    bytes callData;
    uint256 maxFeePerGas;
    bytes signature; // EPERVIER signature
    // ... other fields
}
```

#### Validation:

```
function validateUserOp(UserOperation calldata userOp)
    external returns (uint256 validationData)
{
    bytes32 userOpHash = getUserOpHash(userOp);
    address recovered = EPERVIER.recover(userOpHash,
        userOp.signature);
    if (recovered != address(this)) return
        SIG_VALIDATION_FAILED;
    return 0; // success
}
```

**Gas Budget:** EIP-4337 allocates 4M gas for UserOp validation. EPERVIER’s 1.9M gas fits comfortably, leaving 2.1M for additional logic (e.g., session keys, spending limits).

### 7.3 Native Precompile Proposal

Long-term viability requires native EVM precompile support. We propose:

**Precompile Address:** 0x0b (next available after 0x0a for Blake2f)

**Interface:**

```
Input:  [pubkey_hash (32B)] [signature (666B)] [message (
        var)]
Output: [success (1B)] 0x01 if valid, 0x00 otherwise
Gas:    30,000 (comparable to ECDSA)
```

#### Rationale:

- NTT amenable to hardware acceleration (GPU, FPGA, ASIC)
- Estimated 100-500x speedup vs. Solidity implementation
- Target: <50k gas (20x cheaper than current)
- Precedent: BLS12-381 precompiles (EIP-2537) for Ethereum 2.0

## 8 Comparison with Alternative PQC Schemes

### 8.1 Dilithium

Dilithium [10] is NIST’s primary lattice-based signature standard:

#### Advantages:

- Simpler implementation (no FFT sampling)
- Deterministic signing (no randomness needed)
- Slightly faster verification

**Disadvantages:**

- Larger signatures: Dilithium2 = 2420 bytes (3.6x FALCON-512)
- Larger public keys: 1312 bytes vs. 897 bytes
- Higher bandwidth cost

**EVM Feasibility:**

- Estimated 3-4M gas (2x ETHFALCON)
- NTT also required (same optimization applies)
- No clear advantage for Ethereum deployment

## 8.2 SPHINCS+

SPHINCS+ [11] is a hash-based signature scheme:

**Advantages:**

- Provably secure (minimal assumptions)
- No structured assumptions (no lattices)
- Simple implementation

**Disadvantages:**

- Massive signatures: SPHINCS+-128f = 17,088 bytes
- Slow signing: >100ms for fast variant
- Calldata cost: 68k gas (vs. 2.6k for FALCON)

**EVM Feasibility:**

- Estimated 5-10M gas (verification cost)
- Calldata alone costs 1.1M gas
- Impractical for Ethereum deployment

## 8.3 Summary Comparison

| Scheme        | PubKey | Sig     | EVM Gas | Status    |
|---------------|--------|---------|---------|-----------|
| ECDSA         | 33 B   | 65 B    | 3k      | Current   |
| FALCON-512    | 897 B  | 666 B   | 7M      | This work |
| ETHFALCON     | 897 B  | 666 B   | 1.8M    | This work |
| EPERVIER      | 20 B*  | 2202 B  | 1.9M    | This work |
| Dilithium2    | 1312 B | 2420 B  | 3-4M    | Estimated |
| SPHINCS+-128f | 32 B   | 17088 B | 5-10M   | Estimated |



\* Address only (hash of public key)

**Conclusion:** ETHFALCON/EPERVIER offer the best balance of security, performance, and Ethereum compatibility among NIST PQC candidates.

## 9 Future Work

### 9.1 Hardware Acceleration

NTT operations are highly parallelizable:

- **GPU:** CUDA implementations achieve  $<1\text{ms}$  verification
- **FPGA:** Custom NTT circuits reduce latency to  $<10\mu\text{s}$
- **ASIC:** Dedicated chips (future Ethereum clients)

### 9.2 Signature Aggregation

Lattice-based signatures support aggregation [12]:

- Combine  $n$  FALCON signatures into single proof
- Verification cost grows sublinearly:  $O(n \log n)$  vs.  $O(n)$
- Applications: Rollups, batch transactions, consensus

### 9.3 Threshold Signatures

Multi-party computation for FALCON:

- Distribute secret key across  $n$  parties
- Require  $t$  of  $n$  for valid signature
- Use cases: Multi-sig wallets, DAO governance

### 9.4 Formal Verification

Cryptographic proofs in Coq/Lean:

- Machine-checked security proofs
- Verified implementations (no bugs)
- Integration with formal EVM semantics

## 10 Conclusion

This paper presented ETHFALCON, a suite of EVM-optimized post-quantum signature schemes based on NIST’s FALCON standard. Through strategic cryptographic modifications—replacing SHAKE with Keccak-CTR, implementing iterative NTT, and introducing public key recovery via NTT-domain representations—we achieved 70-90% gas cost reduction from 24M to 1.8-1.9M gas per verification.

Our EPERVIER variant introduces the first FALCON-compatible public key recovery mechanism, enabling quantum-resistant Account Abstraction (EIP-4337, EIP-7702) on Ethereum. With 128-bit quantum security, production-ready implementations, and testnet deployments, ETHFALCON establishes the first operationally viable path to post-quantum blockchain security.

As quantum computing advances, ETHFALCON provides Ethereum with a migration path preserving network security, smart contract compatibility, and user experience. Future work on hardware acceleration, signature aggregation, and native precompile integration will further solidify ETHFALCON’s role in the post-quantum blockchain era.

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